

CUNXIN FAN

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EDUCATION

Shanghai Jiao Tong University

Bachelor of Information Engineering, Bachelor of French

Graduating June 2026

SJTU-ParisTech Elite Institute of Technology

GPA: 4.02/4.30 (92.5/100); **Ranked 2nd /55 (1st /86** in first year)

Relevant coursework: Mathematical Foundations of AI, Probability and Statistics, Topology and Differential Geometry, Neural Networks

Advisor: **Professor Junchi Yan**

University of California, Berkeley

Visiting Student

From July 2024 to May 2025

Advisor: **Professor Masayoshi Tomizuka, Mingyu Ding**

PUBLICATIONS

Interleave-VLA: Enhancing Robot Manipulation with Image-Text Interleaved Instructions

Cunxin Fan*, Xiaosong Jia*, Yihang Sun, Yixiao Wang, Jianglan Wei, Ziyang Gong, Xiangyu Zhao, Masayoshi Tomizuka, Xue Yang, Junchi Yan, Mingyu Ding

ICRA 2025 Workshop Safe-VLM (Spotlight), Under review at ICLR 2025

[\[Dataset\]](#)[\[Code\]](#)[\[Paper\]](#)

WOMD-Reasoning: A Large-Scale Dataset for Interaction Reasoning in Driving

Yiheng Li*, **Cunxin Fan***, Chongjian Ge, Zhihao Zhao, Chenran Li, Chenfeng Xu, Huaxiu Yao, Masayoshi Tomizuka, Bolei Zhou, Chen Tang, Mingyu Ding, Wei Zhan

ICML 2025

[\[Dataset\]](#)[\[Code\]](#)[\[Paper\]](#)

Learning Cryptographic Problems via ANF-based SAT Instance Representation

Xinhao Zheng, Yang Li, **Cunxin Fan**, Huaijin Wu, Xinhao Song, Junchi Yan

NeurIPS 2024

[\[Paper\]](#)

RESEARCH

Research Assistant @ UCB MSCLab, advised by Prof. Masayoshi Tomizuka and Mingyu Ding

Interleave-VLA: Robot Manipulation with Interleaved Instructions – Project Lead

Dec 2024 - May 2025

- Improved **zero-shot generalization** of OpenVLA and π_0 both in simulation and real-world via **multi-modal training** with interleaved image-text instructions.
- Built a **scalable pipeline** to generate interleaved image-text instruction data and extended SimplerEnv.
- Identified **attentional hallucination** in VLAs and demonstrated that Interleave-VLA mitigates it.

WOMD-Reasoning: Interaction Reasoning in Driving – Project Co-Lead

Jul 2024 - Dec 2024

- Developed a **motion-language model combining LLM planning and reasoning with trajectory prediction**, demonstrating its language output enhances motion prediction tasks.
- Explored the application of **CoT** and **DPO** to proposed motion-language models.

Research Assistant @ SJTU RethinkLab, advised by Prof. Junchi Yan and Xiaosong Jia

Representation Matters: VLA Training with External Feature Distillation – Project Lead

Aug 2025 - Present

- Identified **misaligned representations** between action and vision-language tokens in VLAs.
- Demonstrated that external foundation model **feature distillation** outperforms Mixture-of-Experts (MoE) while reducing reliance on explicit labels.

Learning Cryptographic Problems via ANF-based SAT Instance Representation

Apr 2024 - Jun 2024

- Designed **GNN**-based solvers for Algebraic Normal Form SAT problems.
- Validated the approach on solving common cryptographic ciphers: Simon, Speck, and SR.

ROBOTICS COMPETITION

RoboMaster University Championship

Oct 2022 - May 2024

- As **leader of project radar**, developed a localization system in solving front-view **lidar-camera fusion** perception in order to localize all the visible robots.
- Participated in development of an **autonomous navigation** robot, engineered and maintained an autonomous decision-making, localization, and navigation system with ROS.
- Participated in development of a **quadcopter**, engineered and maintained a **PX4-based flight control** system enabling stable hovering via **visual-inertial odometry (VIO)**, along with an auto-aiming module.
- Please refer to [my blog](#) for details.

VEX Robotics Competition (VRC)

Apr 2017 - May 2018

- Implemented C PID controllers for chassis, lift, and swing, improving stability and accuracy.
- Designed a 1-minute fully autonomous routine; ranked 3rd at the World Championship Skill Challenge.

PROJECTS

An Interactive VR Video Game using Motion Capture and Unity

Mar 2023 - Jul 2023

- Developed a VR game featuring real-time motion capture, photorealistic 3D environments using Gaussian splatting, and an LLM-powered dialogue system.

A College New Year Gala Lottery Platform Website

[\[Code\]](#) Nov 2023 - Dec 2023

- Developed a lottery website utilizing Vue3 for frontend, Spring Boot for backend and Mybatis for DAO.
- Incorporated SJTU account login and user avatars; Implemented real-time bullet screen by Websocket.

AWARDS AND HONORS

Academic Honors

- **National Scholarship**, Rank **1st** in college, Top **0.2%** national-wide 2023
- **National Scholarship**, Rank **1st** in college, Top **0.2%** national-wide 2024

Mathematical Modeling

- **Outstanding Winner and SIAM Award, Top 20 of 18,525 (0.1%)**, Team 2425792, Mathematical Contest in Modeling 2024

Competitive Programming

- **Silver**, Asia-Pacific Informatics Olympiad (China) 2020
- **Silver**, Shanghai-China Collegiate Programming Contest 2022
- **First Prize and Top 3%**, National Olympiad in Informatics in Provinces, Shanghai 2019 – 2020
- **Platinum Division**, USA Computing Olympiad 2020

Robotics Competition

- **Champion**, Team Jiao Loong, RoboMaster University Championship, hosted by DJI 2023
Served as project leader of Robot Radar
- **Division Finalist**, Team 8825A, VEX Robotics Competition World Championship 2018
Served as chief programmer
- **Excellence Award**, Team 8825A, VEX Robotics Competition Asia Open 2017
Served as chief programmer

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, C#, C

Software&Frameworks: ROS, PyTorch, TensorFlow, OpenCV, Unity, Blender, T_EX, Typst, MATLAB, Git

Robotics&Hardware: SLAM, Livox LiDAR (Mid-70, Mid-360), Hikvision industrial camera, Realsense depth camera, IMU, NVIDIA Jetson, MoCap, Fanuc robot arm

Languages: Mandarin, English, French